

Nine-Species MEA ePC-SAFT Parameter Bundle

Status: active MEA parameter bundle. This repository owns the live MEA bundle recorded here. A bounded, direct Engine campaign retains a Uyan-style H₂O/MEA mass-fraction dielectric transfer and selects the displayed R4 carbamate-hydrolysis correlation and ϵ_{CO_2}/k . Relative to the previous bundle, the selected result improves full-data pressure $\log_{10}$ RMSE from 0.3974 to 0.3540, temperature-balanced pressure RMSE from 0.3569 to 0.3185, and pressure coverage from 125/161 to 130/161 while retaining all 131/131 observed speciation targets. Value color alone identifies one of the four evidence classes; parameter values use regular weight so typography does not imply a second status system. This PDF is the live present-state view: new accepted evidence replaces the corresponding prior interpretation in place, while Git preserves history.

1. Question and scope

What full parameter bundle is being used for the current calculations? This notebook records the nine-species model structure, pure-component values, association graph, sparse binary interactions, electrostatic choices, reaction correlations, and known non-adopted diagnostics in one reviewable view. Section 9 replays the exact active vector after the audited R1–R3 standard-state correction using its pinned Engine wheel; Section 8 identifies the retained pre-campaign smooth-grid visual. The reported results use the parameter set shown in Tables 1–4.

The liquid species are CO₂, MEA, H₂O, MEAH⁺, MEACOO[−], HCO₃[−], CO₃^{2−}, H₃O⁺, and OH[−]. The neutral vapor species and standard-state details remain governed by the future immutable packet.

1.1. How to read this live notebook

verified statements are checked against the retained input or result files, the current repository, an official software source, or a located primary source. **inference** statements interpret those checked facts. **assumption** identifies a proposed test condition, and **unknown** marks a question that the retained evidence does not answer. Unrun experiments are listed only in Section 12; they are not results.

Evidence	Present state	Meaning
verified	Parameter set	Tables 1–4 contain the active nine-species vector retained here.
verified	Current replay	All 131 observed speciation targets and 130/161 canonical pressure states evaluate with the selected bundle; current full-data metrics are reported in Section 9.
verified	Upstream work	ePC-SAFT Issues #80, #119, #126, and #149 remain open. The native Regression and repository-owner cutovers have merged through upstream PR #169, so Issue #119’s old pre-build blocker text is stale even though the scientific campaign has not run.

2. Model conditions

Choice	Current foundation
Residual Helmholtz terms	PC-SAFT hard-chain and dispersion; general-site association; Debye–Hückel and Born electrostatics.
Association topology	Reciprocal induced CO ₂ -H ₂ O cross association plus ordinary MEA and water association. No CO ₂ self-association and no separate CO ₂ -MEA chemical-association edge.
Permittivity and Born formulation	Uyan-style H ₂ O/MEA mass-fraction transfer: normalized MEA–water mass fractions determine bulk ϵ_r ; dissolved CO ₂ and ions do not enter the mixing rule. Original Born remains active with $c_{\text{shell}} = 0$ and $c_{\text{dielectric}} = 0$.
Parameter identity	Active Uyan-style/original-Born bundle. Parameter SHA-256: 2666914f0f9cfefbdf230e96565de843f9aad9424035c940883147ff66af035c.
Refinement boundary	The campaign changes only R4 and ϵ_{CO_2}/k from the prior bundle. R1–R3, R5, the ion block, association topology, and all binary interactions remain fixed.

3. Complete component parameter bundle

Here m is segment number, σ is segment diameter, ϵ/k is dispersion energy, ϵ^{AB}/k is self-association energy, and κ^{AB} is the stored site-volume coordinate. For CO₂, zero self-association energy disables self-association and the listed site volume is used only by the induced cross-association set. Here z is charge and d_B is Born diameter, ϵ_r is relative permittivity, and f_{solv} is the solvation factor. Lengths are in Å and energies in K. References identify the literature lineage; the underlying exact vector is `results/selected-current-best-parameters.json`. The Debye–Hückel diameter is derived as $d_{\text{DH},i} = 0.88\sigma_i$ and is therefore not tabulated as an independent parameter. The hash in Section 2 identifies the unrounded authority behind the four-decimal-place display.

Table 1. Neutral-component parameters. Only parameters applicable to the neutral molecular rows are shown.

Species	Scheme	m_i	σ_i	ϵ_i/k	ϵ_i^{AB}/k	κ_i^{AB}	$\epsilon_{r,i}$	Ref.
CO ₂	2B ind.	2.0729	2.7852	173.4403	0.0000	0.0451	— ^a	[9, 15, 17]
MEA	2B	3.0353	3.0435	277.1740	2586.3000	0.0375	32.0000	[1, 2, 4]
H ₂ O	2B	1.2047	$\sigma(T)$	353.9500	2425.7000	0.0451	$\epsilon_r(T)$	[8, 11, 16, 17]

^a Go to Section 4.2 for the source, exact derivation, temperature dependence, and replacement decision.**Table 2. Ionic-species parameters.** Only parameters applicable to the ionic rows are shown.

Species	m_i	σ_i	ϵ_i/k	z_i	Ref.
MEAH ⁺	1.0000	3.4851	232.6872	+1	[8, 18, 19]
MEACOO [−]	1.0000	3.5354	453.2652	−1	[18, 19]
HCO ₃ [−]	1.0000	2.9296	70.0000	−1	[12, 17]
CO ₃ ^{2−}	1.0000	2.4422	249.2600	−2	[12, 17]
H ₃ O ⁺	1.0000	3.4654	500.0000	+1	[8, 12]
OH [−]	1.0000	2.0177	650.0000	−1	[8, 12]

Table 3. Active Born and solvation parameters. Ion-specific dielectric-suppression coordinates are not tabulated because the selected solvent-only formulation does not use them.

Species	$d_{B,i}$	$f_{solv,i}$	Ref.
CO ₂	—	1.0000	[8]
MEA	—	1.0000	[8]
H ₂ O	—	1.5000	[8]
MEAH ⁺	3.5332	1.0000	[8, 18, 19]
MEACOO [−]	3.5411	1.0000	[8, 18, 19]
HCO ₃ [−]	3.0000	1.0000	[8, 12]
CO ₃ ^{2−}	3.0000	1.0000	[8, 12]
H ₃ O ⁺	1.2180	1.0000	[8, 12]
OH [−]	3.0811	1.0000	[8, 12]

Table 4. Binary interaction matrix, k_{ij} . Upper triangle shown; the diagonal is structural and unclassified.

Species	CO ₂	MEA	H ₂ O	MEAH ⁺	MEACOO [−]	HCO ₃ [−]	CO ₃ ^{2−}	H ₃ O ⁺	OH [−]
CO ₂	—	0.0000	0.0133	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MEA		—	−0.0735	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
H ₂ O			—	0.0000	0.0000	0.0000	−0.2500	0.2500	−0.2500
MEAH ⁺				—	−0.0020	0.0000	0.0000	0.0000	0.0000
MEACOO [−]					—	0.0000	0.0000	0.0000	0.0000
HCO ₃ [−]						—	0.0000	0.0000	0.0000
CO ₃ ^{2−}							—	0.0000	0.0000
H ₃ O ⁺								—	0.0000
OH [−]									—

Class 1: confirmed current foundation**Class 3:** local fit or derivation awaiting promotion**Class 2:** literature-backed or evidence-informed candidate**Class 4:** missing value, default, or indirect analog

4. Evidence-class justification

The class is assigned to each independent displayed value, not to an entire species. Class 1 is fixed for the current foundation and reopens only when contradictory evidence is admitted. Class 2 is a direct literature value or an explicit, evidence-informed working assumption worth testing for this MEA application. Class 3 is a local fit or derivation with numerical evidence that has not passed promotion checks. Class 4 is a missing independent value represented only by a default or indirect analog. This fourth class was previously collapsed into Class 3 only to follow the earlier request for three classifications; no scientific argument required that collapse. Separating local numerical evidence from truly missing values is more useful for the present campaign.

4.1. Class 1: confirmed current foundation

CO₂. The 2B induced scheme, $m = 2.0729$, and $\sigma = 2.7852 \text{ \AA}$ are the Gross–Sadowski molecular row carried through the induced-association literature. Zero self-association energy and $\kappa^{AB} = 0.0451$ implement the Pabsch–Schick induced topology rather than a fitted CO₂ self-association term [9, 15, 17]. The dispersion energy is no longer Class 1; its promoted local value is documented under Class 3.

MEA. The displayed m , σ , ϵ/k , ϵ^{AB}/k , and κ^{AB} are the Baygi 2B row [4]. The 2B topology is the current selection recorded by upstream PR 132; the older 3B diagnostic is not the retained foundation.

H₂O. The Held 2B family fixes m , $\sigma(T)$, ϵ/k , ϵ^{AB}/k , and κ^{AB} [11, 17]. Figiel et al. directly fit $f_{\text{solv},\text{H}_2\text{O}} = 1.5$ to NaBr mean-ionic-activity-coefficient data [8, Eqns. 8–10 and Tables 2–3].

Neutral-solvent permittivities. The retained water correlation is the Archer–Wang approximation transferred through Uyan’s implementation and gives $\epsilon_{r,\text{H}_2\text{O}}(298.15 \text{ K}) = 78.4104$. This is close to, but not identical with, the high-accuracy 0.1-MPa value 78.3752 from Pátek et al.; the difference 0.0352 exceeds that reference’s stated uncertainty of about 0.01 [16, Eq. 11 and Tables 7–8]. Pátek’s correlation ends at 383.15 K, whereas this diagnostic reaches 393.15 K, so it cannot validate the upper endpoint. The retained $\epsilon_{r,\text{MEA}} = 32.0000$ is also a confirmed current foundation: Afify and Sweatman report an experimental literature average of 32.8 ± 2.1 at 298 K, and the direct dielectric-relaxation measurements of Agieienko and Buchner give 30.45 for neat MEA at 298.15 K [1, Table II] [2, p. 2314]. The displayed 32.0000 is the retained rounded working representative within that evidence range, not an exact measured value or temperature correlation and not a value reported by Baygi and Pahlavanzadeh. Baygi supports the other MEA PC-SAFT coordinates in Table 1 but contains no dielectric measurement or permittivity correlation.

All six ions. One segment, the signed integer charge, and $f_{\text{solv},i} = 1$ are model conventions rather than fitted single-ion properties. Figiel et al. set every pure-ion solvation factor to one because a hypothetical pure-ion system contains no solvent shell [8]. These conventions are Class 1 for this model definition.

Neutral binary interactions. The current $k_{\text{CO}_2,\text{H}_2\text{O}} = 0.0132628791769$ is the accepted constant fit to 39 qualified Kiepe rows with the induced association fixed [14]. The current $k_{\text{MEA},\text{H}_2\text{O}} = -0.0735274987499$ is the accepted 2B fit to 25 admitted Cai rows [7]. Their calculation records and acceptance limits are owned by ePC-SAFT Issue 80; Class 1 here does not make the full nine-species packet accepted.

Protocol-fixed zero corrections. Uyan et al. used the same CO₂, HCO₃[−], CO₃^{2−}, H₃O⁺, and OH[−] species and stated that all binary parameters not separately supplied for the predictive reacting ePC-SAFT calculation were set to zero [18]. Accordingly, the four CO₂–trace-ion corrections and all six trace-ion–trace-ion corrections are Class 1 zeros for the current sparse foundation. Here $k_{ij} = 0$ means that the uncorrected ePC-SAFT combining rule is used; it does not mean that the physical interaction vanishes. These ten values remain outside the single-coordinate CO₂–MEA regression defined by upstream Issue 119.

4.2. Class 2: direct literature candidates

Non-adopted CO₂ permittivity. The source trail is resolved. Schick et al. define CO₂ as a solvent in their composition mixing rule and report

$$\epsilon_{r,\text{CO}_2}(T) = -0.0036T + 2.467, \quad T \text{ in K},$$

as a correlation developed from three source groups [17, Table 2, p. 4]: Leeke et al., the thermophysical report by Ely et al., and the gas-property measurements by Uhlig et al. The historical scalar is exactly this correlation at 293.0000 K:

$$-0.0036(293.0000) + 2.467 = 1.4122.$$

The earliest retained code copy confirms the same lineage with the adjacent comment “Schick 2022”; the paper’s accepted publication year is 2023.

This resolves the numerical origin but does not validate 1.4122 as a temperature-independent property. The Schick correlation gives 1.4117, 1.3937, 1.3397, 1.2677, 1.1957, and 1.1237 at 293.15, 298.15, 313.15, 333.15, 353.15, and 373.15 K, respectively. Leeke et al. measured CO₂ permittivity at 353 and 383 K over 100–300 bar, while independent measurements show that CO₂ permittivity depends on density and pressure as well as temperature. The Schick equation is therefore an ePC-SAFT modeling correlation, not a universal pure-fluid property.

Decision. The full-domain comparison in Section 11.1 selects the Uyan-style solvent-only transfer, which excludes dissolved CO₂ from the dielectric mixing rule. Therefore neither 1.4122 nor the Schick temperature correlation is an active coordinate in Table 1. Both remain retained Class 2 comparison evidence. This decision does not alter the Class 1 molecular Gross–Sadowski row.

SSM+DS neutral solvation factors. Figiel et al. define f_{mix} using ions and solvents only, set pure-ion factors to one, and fit solvent-specific factors of 1.5, 1.4, and 1.6 for water, methanol, and ethanol, respectively [8, Eqns. 8–10 and Table 2]. They also report that 1.5 represents all three studied solvents at low ion concentration. On that evidence, $f_{\text{solv,MEA}} = 1.5$ is the primary proposed value for the first MEA SSM+DS screen: MEA contributes to the solvent environment, is polar and associating, and 1.5 is central to the observed solvent range. The exact bundle evaluated in this notebook still has $f_{\text{solv,MEA}} = 1.0$, so Table 3 now reports 1.0 rather than silently presenting the untested 1.5 proposal as a live value. Both are Class 2 assumptions because no MEA activity data were used to fit either one.

The published mixing rule includes ions and solvents; no other component type was present in Figiel’s systems. Accordingly, $f_{\text{solv,CO}_2} = 1$ is retained as a Class 2 role-based assumption meaning “no neutral-solute shell increment,” not as a published CO₂ property. Its influence should be checked cheaply, but a dedicated fit is low priority unless that sensitivity is material. A current Engine-source inspection finds that the implemented composition average sums all supplied components without a solvent-role mask. Thus the intended solvent-only interpretation and the current implementation must be compared before fitting a CO₂ factor.

Trace-ion molecular rows. The σ_i and ϵ_i/k values for HCO₃[−], CO₃^{2−}, H₃O⁺, and OH[−] are the direct Held/Uyan aqueous-ion rows carried into later ePC-SAFT work [12, 18, 19]. They are credible literature values but have not been independently ranked for reactive aqueous MEA.

Non-adopted ionic dielectric transfers. The former common ionic-region $\epsilon_{r,i} = 8$ and c_i values are direct SSM+DS literature coordinates retained in the comparison provenance, not active MEA-system parameters under the selected solvent-only mixing rule [8, 19]. The H₃O⁺ Born diameter $d_B = 1.2180 \text{ \AA}$ is also directly supported by Figiel et al. [8].

Water–ion interactions. The literature mapping retains $k_{\text{H}_2\text{O,HCO}_3^-} = 0$, $k_{\text{H}_2\text{O,CO}_3^{2-}} = -0.25$, $k_{\text{H}_2\text{O,H}_3\text{O}^+} = 0.25$, and $k_{\text{H}_2\text{O,OH}^-} = -0.25$ [18, 19]. The bicarbonate zero is a literature assumption, not a confirmed physical exclusion; Wangler later fitted nonzero bicarbonate interactions in another electrolyte application. All four therefore remain Class 2 for MEA.

CO₂–MEA. Baygi and Pahlavanzadeh used $k_{\text{CO}_2,\text{MEA}} = 0$ and explicitly interpreted zero as use of the uncorrected PC-SAFT combining rule, not absence of molecular interaction [4]. That exact-species result makes zero the direct literature candidate. It remains Class 2 because their non-electrolyte association model is not the current induced,

nine-species ePC-SAFT foundation. The bounded screen in Section 7 retains zero as the calibration baseline and -0.1957268589 only as a second optimizer start. Neither value is promoted as a fitted result.

Other zero binary corrections. Uyan et al. state that binary parameters not separately supplied for the predictive reacting ePC-SAFT calculation were set to zero [18]. The 18 additional zero entries in Table 4 therefore use the uncorrected combining rule as a literature-backed model convention. They are Class 2 rather than Class 4: the numerical choice is supported, but none is confirmed as an MEA-specific optimum or evidence that the physical interaction vanishes.

4.3. Class 3: project-fitted or derived values

The current $\epsilon_{\text{CO}_2}/k = 173.440\,25\text{ K}$ is a 2.5% increase from the Gross–Sadowski start. It was selected jointly with R4 by the bounded campaign in Section 7. Both coordinates are Class 3 because they are project-fit values rather than values copied from a paper.

The May 2026 local-speciation fit varied seven coordinates simultaneously: σ_i , ϵ_i/k , and $d_{\text{B},i}$ for both MEA^+ and MEACOO^- , plus $k_{\text{MEA}^+, \text{MEACOO}^-}$. It used 8 states and 22 residuals from Böttinger, Jakobsen, and Matin data. Overall log-residual RMSE changed only from 0.2714 to 0.2677, and two final evaluations failed. Accordingly, the six displayed ion parameters and the -0.0020 interaction are Class 3 because they are project-fit outputs. Git commit 897bb6e preserves the exact fit values and diagnostics.

The OH^- Born diameter $d_{\text{B}} = 3.0811\text{ Å}$ is a local Born hydration-energy inversion retained under `analyses/reactive_epcsaft_parameter_evidence/results/oh_born_derivation/`. It remains Class 3 until its source assumptions and transfer checks are accepted. Every Debye–Hückel diameter is excluded from the tables and this classification inventory because it is deterministically calculated as $0.88\sigma_i$.

4.4. Class 4: missing values represented by defaults or analogs

The HCO_3^- and CO_3^{2-} Born diameters of 3.0000 Å are Class 4 seeds. The retained trace-carbonate check left both values at their starts with no residual improvement, so it did not convert the defaults into fitted parameters. The former $c_{\text{MEA}^+} = 2.6000$ and $c_{\text{MEACOO}^-} = 7.0100$ defaults are retained only as non-adopted SSM+DS provenance; the active bundle does not use them.

The matrix diagonal is gray and unclassified because self-pairs do not use a binary interaction coefficient.

5. Induced-association lineage and correlations

The CO_2 molecular row is not a new induced-association fit. Gross and Sadowski report $m = 2.0729$, $\sigma = 2.7852\text{ Å}$, and $\epsilon/k = 169.21\text{ K}$ in their original PC-SAFT table [9]. Pabsch et al. retain those values, add a 1/1 induced 2B site topology with zero CO_2 self-association energy and $\kappa^{AB} = 0.0450$, and fit the CO_2 –water binary term [15, Tables 2 and 6]. Schick et al. carry the same induced row forward [17, Tables 1 and 2]. The retired $m = 2.079$ value is the unsupported value; 2.0729 is the source-correct Gross–Sadowski value.

The retained water correlations are

$$\sigma_{\text{H}_2\text{O}}(T) = 2.7927 + 10.11 \exp(-0.01775T) - 1.417 \exp(-0.01146T), \quad (1)$$

$$\epsilon_{r,\text{H}_2\text{O}}(T) = 78.41043478578715 - 0.36133766233691506(T - 298.15) \quad (2)$$

$$+ 0.00076555618295(T - 298.15)^2. \quad (3)$$

The retained CO_2 scalar is a 293 K snapshot of the Schick correlation $\epsilon_{r,\text{CO}_2}(T) = -0.0036T + 2.467$, not a constant supported across the calculation range [17, Table 2, p. 4]; see Section 4.2.

The current bundle uses the selected **reciprocal induced CO_2 – H_2O association set**. CO_2 , MEA, and H_2O each carry one donor site a and one acceptor site b . Upstream PR #132 independently selected MEA 2B over 3B and 4C using the

admitted Cai MEA–water rows. The CO₂–water interaction is physical; reactive carbamate formation remains in the reaction network.

Pair	Active edges	ϵ^{AB}/k (K)	κ^{AB}	Meaning
MEA–MEA	$a:b, b:a$	2586.30	0.03747	Ordinary MEA self-association.
H ₂ O–H ₂ O	$a:b, b:a$	2425.70	0.04509	Ordinary water self-association.
CO ₂ –H ₂ O	reciprocal a – b cross edges	1212.85	0.04509	Selected induced association: CO ₂ (a)–H ₂ O(b) and CO ₂ (b)–H ₂ O(a).
MEA–H ₂ O	reciprocal a – b	2506.00	0.0411037990944876	Combined cross-association rule.
CO ₂ –CO ₂	none	—	—	Self-association disabled.

6. Reaction correlations

For R1–R3, the Austgen source form is $\ln K_{r,\text{source}} = A + B/T + C \ln T + DT$. Before constructing an Engine problem, the replay evaluates

$$\ln K_{r,\text{common}}(T) = \ln K_{r,\text{source}}(T) + \Delta_r, \quad r \in \{\text{R1}, \text{R2}, \text{R3}\}, \quad (4)$$

with $(\Delta_1, \Delta_2, \Delta_3) = (8.0330699846, 4.0165349923, 4.0165349923)$. These are the audited shifts from the Austgen mixed mole-fraction reference to aqueous–molality–infinite–dilution–water–v1. R4 and R5 already retain their declared common–molality forms. All five reactions remain fixed while physical parameter blocks are qualified.

ID	A	B	C	D	Domain (K)	Source definition
R1	132.899	–13445.9	–22.4773	0	273.15–498.15	Water autoprotolysis.
R2	231.465	–12092.1	–36.7816	0	273.15–498.15	CO ₂ hydration to bicarbonate.
R3	216.049	–12431.7	–35.4819	0	273.15–498.15	Bicarbonate dissociation.
R4	$\ln K = 3.3515778 - 1895.3/T$			293.15–393.15 ^a		Carbamate hydrolysis; joint fit.
R5	$\ln K = -\ln(10)[2677.91/T + 0.3869 + 0.0004277T]$			273.15–323.15		MEA ⁺ acid dissociation.

^aThe source correlation was qualified over 293.15–323.15 K. The displayed fit evaluates it through 393.15 K. R2 uses Austgen’s printed $A = 231.465$, which is also the value encoded in the validated state packet; the later 231.456 transcription is retained only as a documented source discrepancy.

7. Corrected replay and parameter-start decision

7.1. Verified packet-construction defect

The upstream reaction audit records the shifts in Eq. (4), but the retained exporter evaluates the raw Austgen R1–R3 correlations and writes those values into records already labeled with the common aqueous–molality standard state. The immutable state packet is preserved unchanged; the notebook replay applies the three shifts before the Engine mapping is constructed. The affected file and Engine commit are identified in Section 12. This distinguishes a packet-export error from a reaction-model or Engine-transfer error.

7.2. Prior ion-specific baseline and reaction screen

The corrected former ion-specific foundation evaluates every one of the 44 speciation states and 76 of the 79 pressure states. Material balances, reaction affinities, and pressure residuals remain below 3.0×10^{-10} in their reported infinity norms. The model therefore solves the stated equilibrium problem accurately where it evaluates; the comparison with observations tests model validity, not nonlinear-solver convergence.

Candidate	N_p	median $p_{\text{model}}/p_{\text{data}}$	pressure RMS factor	N_x	species MAE	species RMS factor
Current R4/R5, corrected R1–R3	76/79	13.89	16.71	44/44	0.00257	1.94
Böttinger R4, corrected R1–R3	70/79	10.25	13.27	44/44	—	2.96

The former ion-specific foundation reproduces the measured species pattern within a median multiplicative factor of 1.065, but its pressure is high by a median factor of 13.89. Böttinger’s common-molality R4 lowers pressure by only 15% on the 70 paired evaluable rows, worsens species agreement, and adds six pressure non-evaluations. Its R5 is numerically indistinguishable from Bates–Pinching over the tested range (less than 1.8×10^{-5} difference in $\ln K$). Neither reaction substitution resolves the pressure discrepancy.

The pressure bias is also temperature structured:

T (K)	313.15	333.15	353.15	373.15	393.15
median $p_{\text{model}}/p_{\text{data}}$	26.75	23.71	7.27	11.03	6.82

7.3. Bounded physical-interaction screens

The constant binary screens used 18 pressure states common to every candidate and six representative speciation states spanning the available temperatures.

Coordinate and value	median pressure factor versus current	species scaled RMSE	Interpretation
$k_{\text{CO}_2, \text{MEA}} = -0.1957268589$	0.9978	0.2536	Retain only as Issue #119’s second optimizer start.
$k_{\text{CO}_2, \text{MEA}} = +0.1015997042$	1.0010	0.2542	Old fit used the unconverted reaction packet; not transferable.
$k_{\text{CO}_2, \text{MEA}} = -1$	0.9841	0.2554	Diagnostic bound remains too weak and adds a species non-evaluation.
$k_{\text{CO}_2, \text{H}_2\text{O}} = 0$	0.7838	unchanged	Still leaves a 7.76 median pressure ratio and conflicts with the independent CO_2 –water qualification.
$k_{\text{CO}_2, \text{H}_2\text{O}} = 0.016724$	1.0661	unchanged	Pabsch’s 313 K value worsens reactive pressure relative to the current constant fit.

The verified conclusion from this earlier bounded screen was that no retained or directly transferable literature bundle presently gives both a reasonable nine-species speciation replay and a reasonable CO_2 pressure replay under the selected explicit-ion structure. That screen therefore defined the then-defensible calibration start as the corrected R1–R3 basis, Tong–Aroua R4, Bates–Pinching R5, and $k_{\text{CO}_2, \text{MEA}} = 0$; the legacy negative value was only a second start. Section 11.1 superseded its dielectric choice, and Section 7.4 supersedes its R4 and CO_2 dispersion coordinates.

Baygi’s reported pressure performance cannot be transferred by copying its parameters. That calculation used 3B MEA, 4C water, nonassociating CO_2 , ideal Smith–Missen chemistry, and $k_{\text{MEA}, \text{H}_2\text{O}} = -0.052$ [4]. The present foundation uses 2B MEA, Held 2B water, induced CO_2 –water association, explicit electrolyte activities, and $k_{\text{MEA}, \text{H}_2\text{O}} = -0.0735275$. The two parameter sets do not represent the same model.

7.4. Bounded joint R4– CO_2 campaign and selection

The September 2026 campaign used direct coupled Engine reactive bubble-point and homogeneous-liquid equilibrium solves throughout. A native-regression smoke test on six states and two coordinates required 129 seconds for only two iterations and did not produce a usable fit, so the retained study used parallel, bounded direct evaluations rather than an ad hoc SciPy equilibrium surrogate. The sequence comprised a centered sensitivity screen, R4 pivot–slope grids, complete-data validation, a local R4– CO_2 refinement, and a final slope-boundary check. No numerical score was treated as an accept/reject gate; candidates were compared on error, convergence coverage, temperature residual pattern, and speciation tradeoffs.

The centered screen found no numerical response to the inactive $f_{\text{solv}, \text{HCO}_3^-}$ coordinate, negligible benefit from the tested bicarbonate Born movement, and worse pressure/speciation tradeoffs from the tested R5 direction. R4 was the only consistently useful reaction

block. Reparameterizing R4 around 333.15 K separated its local pivot from its temperature slope; a small CO₂ dispersion-energy increase then reduced the remaining common pressure bias.

The selected current bundle is

$$\ln K_{R4} = 3.3515778178 - 1895.3/T, \quad \epsilon_{\text{CO}_2}/k = 173.440\,25\text{ K}.$$

All other coordinates remain equal to the previous bundle. The final comparison is:

Bundle	pressure coverage	pressure log RMSE	balanced pressure RMSE	speciation coverage / log RMSE
Previous	125/161	0.3974	0.3569	131/131 / 0.2664
Selected	130/161	0.3540	0.3185	131/131 / 0.2658

Pressure log RMSE improves at every temperature: 0.5047 to 0.4445 at 40 °C, 0.2888 to 0.2683 at 60 °C, 0.2491 to 0.2314 at 80 °C, 0.2315 to 0.1971 at 100 °C, and 0.4280 to 0.3809 at 120 °C. The sign pattern remains diagnostic: mean log residuals are +0.4014 and +0.1698 at 40 and 60 °C, then −0.1052, −0.0552, and −0.3661 at 80, 100, and 120 °C. Thus the current model still overpredicts low-temperature pressure and underpredicts high-temperature pressure. Extending the R4 slope to its present bound did not improve the screen and began losing speciation evaluations, so the bounded direction is closed rather than extrapolated farther.

A separate SciPy reaction-extent spot check at v1e_obs_0128, using Engine composition derivatives at the Engine bubble pressure, returned to the Engine liquid composition within 3.31×10^{-13} in mole fraction and a maximum reaction-affinity residual of 3.31×10^{-8} . This verifies that spot-check root only; no SciPy-derived parameter score is retained.

8. Nine-species equilibrium replay

Figure 1 retains the pre-campaign smooth-grid visual baseline generated with the corrected replay and retained PR #134 state packet. The bounded campaign did not rerun the additional smooth grid because the 131 observed targets were sufficient for candidate selection; exact selected-bundle statistics follow in Section 9. The single 20 °C, 30 wt% MEA panel is the densest retained same-condition comparison: 149 positive values for the five principal species and the MEA+MEA⁺ observation operator from 39 Böttinger, Jakobsen, and Matin states. These include direct, aggregate, inferred, and correlated values. The 17 exact reported-zero records remain in the retained observation snapshot but are omitted from the logarithmic plot. The broken logarithmic scale gives most of the panel to the principal-species range from 5.6248×10^{-5} , 30% below the lowest positive observed or modeled HCO₃[−] value, to 0.1134, only 3% above the displayed maximum. A compact 10^{-10} to that cutoff trace panel retains the low CO₂ curve. No retained positive CO₂ or HCO₃[−] observation is below 10^{-10} ; the values shown there are calculated molecular CO₂, CO₃^{2−}, H₃O⁺, and OH[−] remain in every equilibrium calculation; the eight reported carbonate values also remain in the figure source snapshot, but trace species are omitted from the visual. Dashed curves connect direct pinned-Engine evaluations from loading 0.075 to 1.0; they are not fitted or interpolated curves.

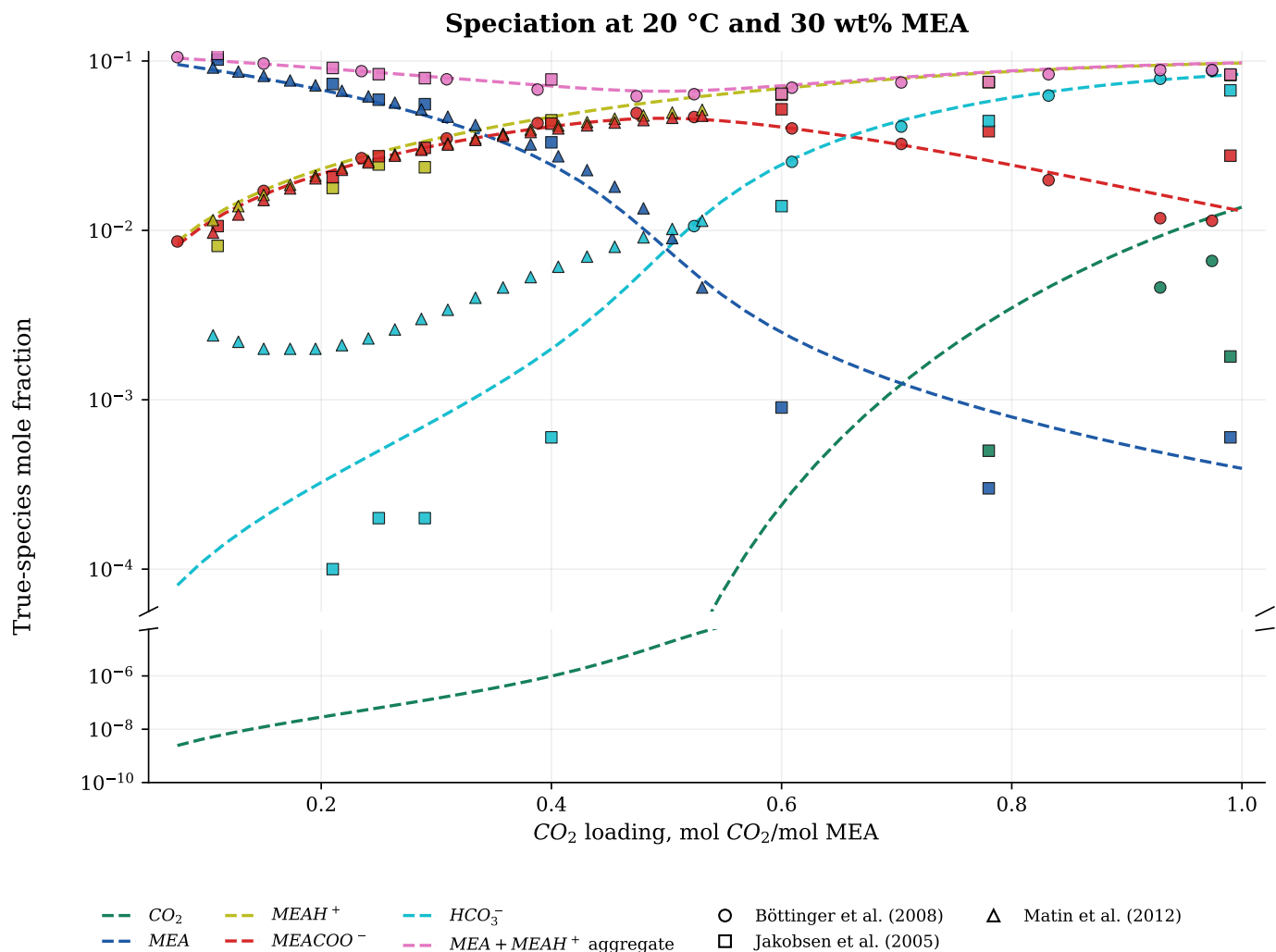


Figure 1. Pre-campaign smooth-grid diagnostic for the full nine-species liquid-equilibrium replay at 20 °C and 30 wt% MEA. Symbols show all positive retained Böttinger, Jakobsen, and Matin values for the displayed series; symbol shape identifies source and color identifies species. The broken logarithmic axis expands the 5.6248×10^{-5} – 0.1134 principal-species region and compresses the 10^{-10} – 5.6248×10^{-5} range. Dashed lines connect direct Engine evaluations of the immediately preceding bundle through loading 1.0; they are retained for qualitative context and are not the authority for the selected- bundle statistics.

These states are homogeneous reactive-liquid solves, not ideal-reaction or postprocessed SciPy curves. Each state couples the five reaction constraints to ePC-SAFT chemical potentials for all nine species at fixed temperature and pressure. A 2026-09-01 disposable timing check on this machine, excluding process startup and model construction, measured 0.935 s/state for a warm five-state second pass with the retained wheel. The same five feeds took 2.260 s/state with a disposable wheel built from Engine main after only the required provider-to-EOS request-schema renames. This is a route-specific diagnostic, not a general Engine benchmark: recent GREPE and regression speedups do not establish a speedup for this homogeneous-liquid owner.

9. Carbon-dioxide pressure replay

Figure 2 shows all 161 active-v1 30 wt% MEA pressure observations from Aronu, Hilliard, Idris, Jou, Mamun, and Xu at 40–120 °C. The selected replay evaluated 130 states and returned 31 explicit numerical convergence failures. Its full-data $\log_{10}$ RMSE is 0.3540 and the equal-temperature value is 0.3185. Lines interpolate the 130 evaluated selected-bundle observation states; no additional smooth-grid Engine campaign was required, and every experimental point remains shown.

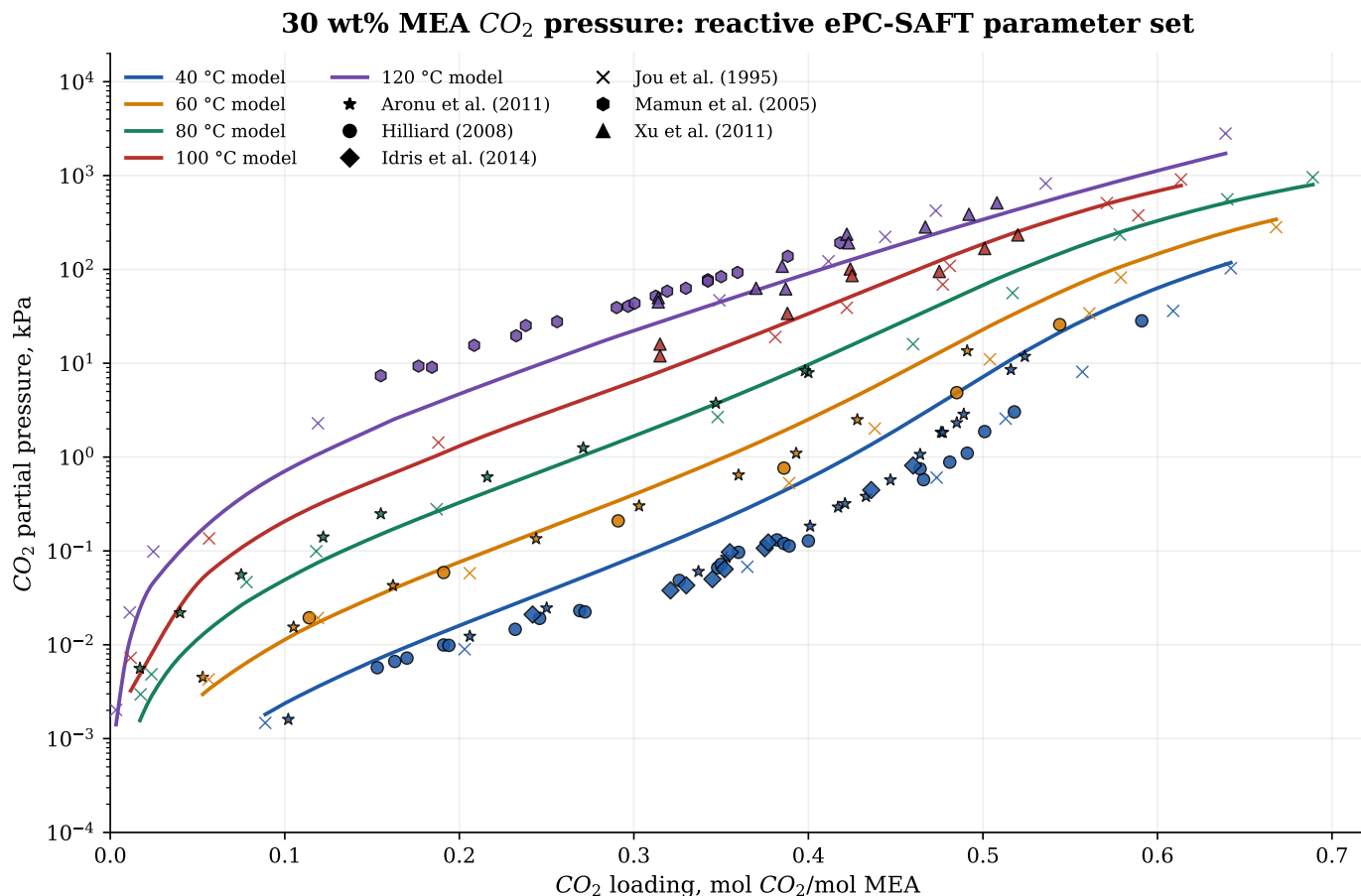


Figure 2. Carbon-dioxide partial pressure for 30 wt% MEA. Colors identify temperature, symbols identify experimental source, and lines are shape-preserving log-response interpolations of newly calculated active-bundle states within each evaluated loading range; every retained observation remains visible even when its corresponding model state did not evaluate.

9.1. Current-bundle fit and coverage statistics

These are replay/agreement statistics for the exact bundle in Tables 1–4. Log residuals are $\log_{10}(y_{\text{model}}/y_{\text{obs}})$; the RMSE factor is $10^{\text{RMSE}_{\log_{10}}}$. Native pressure errors are in kPa and native speciation errors are mole fractions.

Family	Coverage	log RMSE	RMSE factor	MAPE (%)	MAE	Native RMSE
Pressure	130/161	0.3540	2.259	105.04	18.6330	99.9354
Speciation targets	131/131	0.2658	1.844	16.88	0.002643	0.003426

Pressure performance and, especially, evaluability vary strongly by temperature:

T (°C)	Coverage	log RMSE	RMSE factor	MAPE (%)	Median APE (%)
40	56/59	0.4445	2.783	176.41	178.56
60	23/24	0.2683	1.855	75.68	38.99
80	21/21	0.2314	1.704	35.43	38.26
100	18/18	0.1971	1.574	34.71	29.15
120	12/39	0.3809	2.404	55.63	59.32

The principal measured-species breakdown shows that bicarbonate, not the amine/carbamate coordinates, dominates the speciation log score:

Target	Coverage	log RMSE	RMSE factor	MAPE (%)
MEA	19/19	0.0575	1.141	10.07
MEA H^+	19/19	0.0498	1.121	11.38
MEA + MEA H^+	25/25	0.0221	1.052	3.36
MEACOO $^-$	44/44	0.0317	1.076	6.07
HCO $_3^-$	24/24	0.6154	4.125	60.53

The complete machine-readable handoff is retained in the results directory:

- row-level observed/predicted values: `current-best-fit-residuals.csv`;
- grouped overall, temperature, source, species, and temperature–source statistics: `current-best-fit-statistics.csv`;
- failed coordinates and diagnostics: `figure-calculation-failures.csv`.

The bundle can be used for absorption-column calculations. The caller should retain explicit non-evaluation for states that do not solve; at 120 °C, 12/39 pressure states evaluated.

10. What pressure constrains and what remains unresolved

At phase equilibrium the molecular carbon-dioxide fugacities satisfy

$$f_{CO_2}^V = f_{CO_2}^L. \quad (5)$$

An activity/Henry representation writes the same comparison schematically as

$$y_{CO_2} \phi_{CO_2}^V P = x_{CO_2} \gamma_{CO_2}^L H_{CO_2}. \quad (6)$$

inference: pressure observations therefore constrain the liquid molecular- CO_2 chemical potential together with loading, material balances, and the reaction network. They are valuable indirect speciation evidence, but they do not separately identify free x_{CO_2} , liquid nonideality, reaction constants, association, binary interactions, dielectric terms, or solvation factors. Direct species observations and pressure observations are combined because they constrain different projections of the same equilibrium state; pressure is not a substitute for direct speciation.

verified: the retained replay does not assume an ideal vapor. Its state packet declares a neutral incipient-vapor phase and reads `phase.partial_pressure` from the pinned ePC-SAFT equilibrium result. By contrast, the current public IDAES column property files declare an ideal liquid/Henry-law MEA solvent and an ideal vapor; they are not the modified-eNRTL model from a separate literature study. The public solvent file retains the three apparent neutral components CO_2 , MEA, and H_2O plus the three ionic species MEA^+ , MEACOO^- , and HCO_3^- [official IDAES solvent file]; the vapor file explicitly selects the ideal equation of state [official IDAES vapor file].

10.1. Reduced-chemistry diagnostic

The six-species reduction retains CO_2 , MEA, H_2O , MEA^+ , MEACOO^- , and HCO_3^- . To isolate species-set structure without importing IDAES empirical equilibrium constants, derive its two net constants from the current five-reaction basis:



The first relation is the exact R2–R4–R5 projection already recorded in MEA Issue #72. This makes the comparison a reaction-space projection of the same model rather than a change of both chemistry and equilibrium constants.

verified: across the 44 retained nine-species calculations, the largest mole fractions are 1.0307×10^{-2} for HCO_3^- , 1.1687×10^{-4} for CO_3^{2-} , 3.3641×10^{-6} for OH^- , and 8.9480×10^{-12} for H_3O^+ . The largest omitted net-charge contribution $(2x_{\text{CO}_3^{2-}} + x_{\text{OH}^-} - x_{\text{H}_3\text{O}^+})/x_{\text{MEA}^+}$ is 0.01949 at Böttinger state 030. These values are calculated directly from `figures/speciation/output/speciation-model.csv`.

The complete matched comparison is retained under `../enrtl_six_species_ideal_comparison/`; its Quarto source and rendered PDF are the detailed calculation record. In its ideal-liquid lane, all 232 unique canonical coordinates and 464 model rows converged with a maximum equation residual of 1.9895×10^{-13} . The largest difference among the six shared species was 6.5100×10^{-3} mol per mol initial MEA, and carbonate dominated the omitted inventory. In its fixed- T , P Engine/SciPy lane at 100 kPa, the carbonate amounts at 293.15, 313.15, and 323.15 K were 2.99×10^{-4} , 1.56×10^{-4} , and 1.10×10^{-4} mol per mol initial MEA; the largest shared-species changes were approximately 2.9×10^{-4} mol per mol initial MEA. That lane is a liquid-composition calculation, not a bubble-pressure calculation.

verified structural result: when the six- and nine-species states are both projected to the same conserved apparent totals, their neutral apparent pressure inputs agree to 1.321165×10^{-14} . The neutral `flashTQ` route converged for only 1 of 161 pressure observations and predicted 2038.38 kPa for an observed 0.002 02 kPa. A neutral apparent flash therefore cannot reveal the internal reaction or ionic parameters; the pressure calculation must keep the reactive species allocation coupled to the EOS.

The companion sensitivity survey evaluated 143 parameter identities grouped into 113 continuous directions with 128 Latin-hypercube samples, of which 123 were usable. Its strongest recurring directions were R5 a_K , R4 b_K , the water segment-diameter temperature terms, MEA^+ dispersion and ionic terms, and selected CO_2 –water, carbonate–water, MEA– MEA^+ , MEA^+ –water, carbamate–hydronium, and bicarbonate– CO_2 binary interactions. Those numerical ranks belong to parameter packet `0e7c5ac63c078be9d32c15f316ecdb2b0de67de6f495358ab5955450f0271c6a` and Engine-wheel hash `05cca73463b9b44dcb3a77dfd239b767ec63aafce7d18b85b2a66b80773847a5`; the separate fixed- T , P lane used packet `12e380a27eb615e25bae580f8c1ec852020879917b08fa16a5821ae76311550d`. Neither is the active bundle in Tables 1–4, so the ranking must be replayed with the active parameter file before its numerical ordering is used in the manuscript.

inference: the three omitted trace ions have small direct abundance over this replay, which makes a six-species ablation useful as a cheap conditioning test. It does not prove that they are dispensable: the z_i^2 weighting gives carbonate four times the per-mole ionic-strength contribution of a monovalent ion, and the omitted species preserve carbonate, autoprotolysis, pH, charge, and extrapolation structure. Removing them also removes weakly identified ionic parameters, so an apparent fit improvement may come from reduced flexibility rather than better physics.

Retained finding	Present use	Remaining action
Exact R2–R4–R5 and R2–R5 projection	Methods description for a controlled six-species ablation	None for the algebra; cite the retained reaction table.
Ideal and fixed- T , P species comparison	Evidence that carbonate is the dominant omitted species and that the reduction is not simple deletion	Repeat the coupled bubble calculation with the active bundle before using pressure-error conclusions.
Neutral apparent-pressure invariance	Methods/diagnostic explanation of why total absorbed carbon cannot be passed directly to a neutral CO_2 flash	Keep as a route-rejection result; do not present the 1/161 flash as a model fit.
Packet-specific sensitivity survey	Prioritizes the first active-bundle sensitivity directions	Replay the same screen with the active parameter mapping and wheel before reporting ranks or intervals.

11. Interpretation

The R4– CO_2 update reduces pressure error while preserving complete observed- speciation coverage, but the residual changes sign with temperature and 120 °C pressure coverage remains only 12/39. This identifies a remaining temperature-dependent mismatch; it does not justify opening R1–R3, R5, or the ion block as compensating coordinates.

The Class 2 $k_{\text{CO}_2, \text{MEA}} = 0$ is a direct literature candidate for the uncorrected combining rule, not a fitted result or a confirmed absence of interaction. The bounded screen shows that one constant correction is too weak and too temperature uneven to explain the observations. Likewise, loaded-MEA pressure alone cannot identify $f_{\text{solv}, \text{CO}_2}$ or $f_{\text{solv}, \text{MEA}}$ because those working assumptions are confounded with cross association, dielectric response, reaction, and other physical interactions.

Historical diagnostics remain outside the bundle:

- legacy $k_{\text{CO}_2, \text{MEA}} = -0.1957268589$;
- alternate MEAH^+ $\sigma = 2.753332853$ and MEACOO^- $\sigma = 3.017079292$; and
- trace-only bicarbonate Born-diameter movement toward 6.802 94 Å, with CO_3^{2-} left near 2.997 44 Å.

These values identify sensitivity or lineage; none is adopted. The trace-carbonate result does not establish independent identifiability.

11.1. Full-domain permittivity selection

This fixed-parameter structure study asks whether the current Uyan-style solvent-only transfer with original Born remains the best available combined description of reactive MEA pressure, speciation, dielectric response, and physical branch identity. It changes no reaction, molecular, ionic, association, binary, vapor, tolerance, or initialization input. Reciprocal induced CO_2 – H_2O association and the unique Born diameters in Table 2 remain active in every calculation. No parameter was fitted.

11.1.1. Equations, pools, units, and derivatives

Let $N = \{\text{H}_2\text{O}, \text{MEA}, \text{CO}_2\}$, I be the six charged species in Table 2, $x_N = \sum_{j \in N} x_j$, and $w_j^N = x_j M_j / \sum_{k \in N} x_k M_k$. Rule A intentionally uses $N_A = \{\text{H}_2\text{O}, \text{MEA}\}$ because Rule A transfers Uyan et al.’s mixing form; Rules B–E include molecular CO_2 explicitly. No rule silently removes or renormalizes another component. The screened bulk relative-permittivity

rules are

$$\epsilon_r^A = \sum_{j \in N_A} w_j^N \epsilon_{r,j}, \quad \text{Uyan-style solvent-only transfer,} \quad (9)$$

$$\epsilon_r^B = \sum_{j \in N \cup I} x_j \epsilon_{r,j}, \quad \text{Bülow mole-fraction mixing,} \quad (10)$$

$$\epsilon_r^C = \frac{\sum_{j \in N \cup I} x_j M_j \epsilon_{r,j}}{\sum_{j \in N \cup I} x_j M_j}, \quad \text{Ascani mass-fraction mixing,} \quad (11)$$

$$\epsilon_r^D = x_N \sum_{j \in N} w_j^N \epsilon_{r,j} + \sum_{i \in I} x_i \epsilon_{r,i}, \quad \text{Ascani combined rule,} \quad (12)$$

$$\epsilon_r^E = \frac{\epsilon_{r,N}^{\text{salt-free}}}{1 + 7.01 x_I}, \quad x_I = \sum_{i \in I} x_i, \quad \text{Figiel nonlinear rule.} \quad (13)$$

Rules B–D use $\epsilon_{r,i} = 8$ for every ion. The pure-component inputs are the retained water correlation, $\epsilon_{r,\text{MEA}} = 32$, and the Schick transfer $\epsilon_{r,\text{CO}_2}(T) = 1.4122 - 0.0036(T - 293 \text{ K})$ [17, Table 2]. Equations 10–12 follow Bülow and Ascani’s linear component formulations [3, 5, 6]; Equation 13 is Figiel’s empirical electrolyte suppression fitted at 298.15 K [8, Eqns. 11–12]. Relative permittivity, f_{solv} , c_{shell} , and $c_{\text{dielectric}}$ are dimensionless; d_i^{Born} is stored in Å and converted to metres. The exact source locators are Bülow’s original-Born Eq. 2 and composition-derivative Eqs. 3–5, with the linear dielectric discussion around Eq. 7; the Bülow corrigendum’s correction to the input used in Figs. 6–7; Ascani’s Born Eq. 10 and mixing Eqs. 11–13; and Figiel’s SSM+DS Eqs. 6–10, nonlinear dielectric Eqs. 11–12, and Supporting Information Table 1, Table S2, and Figs. S1–S6. The corrigendum reports that the corrected figures do not change Bülow’s conclusions. The full receipt records separate SHA-256 identities for the Bülow article, its corrigendum, the Ascani article, the Figiel article, and the Figiel Supporting Information rather than treating any pair as one source.

“Uyan” here is not an MEA parameter set. Uyan et al. studied aqueous MDEA and used their Eq. 5 salt-free solvent-only mass-fraction average of water and MDEA permittivities [18]. Rule A transfers only that algebraic form to water–MEA and substitutes the retained MEA scalar. Dissolved molecular CO_2 and all ions therefore have zero direct weight in Rule A. Calling it “Uyan MEA” would be inaccurate; throughout this section it means “Uyan-style $\text{H}_2\text{O}/\text{MEA}$ mass-fraction transfer.”

For rules with temperature-dependent pure-component values, the exact Engine automatic derivatives follow the declared algebra. For example, $\partial \epsilon_r^B / \partial x_i = \epsilon_{r,i}$ and, if $\epsilon_r^C = N_C / M_C$, $\partial \epsilon_r^C / \partial x_i = M_i(\epsilon_{r,i} M_C - N_C) / M_C^2$. The corresponding temperature derivatives sum the pure-component $\partial \epsilon_{r,j} / \partial T$ terms; all five rules have $\partial \epsilon_r / \partial \rho = 0$. Rules D and E use exact quotient and chain derivatives rather than frozen-composition approximations.

The original Born residual is

$$\frac{a^{\text{Born}}}{RT} = -\frac{e^2}{4\pi\epsilon_0 k_B T} \sum_{i \in I} x_i z_i^2 \frac{1 - 1/\epsilon_r}{d_i^{\text{Born}}}. \quad (14)$$

The corrected Engine SSM+DS form first defines

$$d_i^{\text{out}} = d_i^{\text{Born}} \left[1 + c_{\text{shell}} \frac{f_{\text{mix}} - 1}{|z_i|} \right], \quad f_{\text{mix}} = \frac{\sum_{j \in S} x_j f_{\text{solv},j}}{\sum_{j \in S} x_j}, \quad (15)$$

and replaces the radial factor in Equation 14 by

$$\frac{1 - 1/\epsilon_r}{d_i^{\text{out}}} + c_{\text{dielectric}} \left(1 - \frac{1}{\epsilon_{r,i}^{\text{local}}} \right) \left(\frac{1}{d_i^{\text{Born}}} - \frac{1}{d_i^{\text{out}}} \right), \quad \epsilon_{r,i}^{\text{local}} = 8. \quad (16)$$

Original Born sets both switches to zero; corrected SSM+DS sets both to one. S is the explicit solvation pool: every component with a declared $f_{\text{solv},j}$. The primary Figiel calculations include all nine species in S ; the coupled CO_2 -exclusion calculation omits molecular CO_2 from both S and the salt-free neutral-permittivity pool. Thus “ CO_2 excluded” means both that CO_2 is not treated as a solvent-like contributor to f_{mix} and that it does not contribute to the salt-free neutral dielectric average; CO_2 remains a full reacting EOS component. No composition is silently renormalized outside the denominators shown above. Equation 16 is the Engine’s corrected nested radial charging expression, not the misprinted ordering in

Figiel’s printed Eq. 7. The Engine differentiates Equations 14– 16 with respect to temperature, density, composition, and active parameters.

The six fixed MEA-study Born diameters are 3.5332 Å (MEA⁺), 3.5411 Å (MEACOO[−]), 3.0000 Å (HCO₃[−]), 3.0000 Å (CO₃^{2−}), 1.2180 Å (H₃O⁺), and 3.0811 Å (OH[−]). They were held fixed in every structure comparison. Figiel et al. instead fitted aqueous infinite-dilution solvation data for H⁺, Li⁺, Na⁺, K⁺, VO₂⁺, V³⁺, Cl[−], Br[−], I[−], and SO₄^{2−}; their Table 3 diameters are respectively 1.218, 2.784, 3.445, 4.150, 2.930, 2.940, 4.100, 4.480, 4.985, and 4.953 Å [8]. Only the H₃O⁺/H⁺ value in the present bundle is an exact overlap. Figiel does not supply diameters for MEAH⁺, MEACOO[−], HCO₃[−], CO₃^{2−}, or OH[−]; substituting Na, halide, or sulfate values for those chemically different ions would not be source-consistent. A later diameter regression must therefore use ion solvation or other ion-specific evidence, not arbitrary cross-ion substitution.

The “ion-specific” α_i screen also needs qualification. Zuber et al.’s Table 2 directly gives 9.55 for H⁺ and 13.96 for OH[−], which map exactly to the present H₃O⁺ and OH[−] records [21]. The present values 2.60 for MEAH⁺, 7.01 for MEACOO[−], and 7.89 for each carbonate species are project-declared analogs or fallbacks; Zuber did not fit those MEA-system ions. Consequently I-SSM+DS is an analog/fallback sensitivity, not a complete literature-matched ion-specific package.

The retained derivative audit independently perturbed one interior composition direction for Rules B and C. Analytic and centered finite- difference derivatives agree to relative errors of 2.21×10^{-11} and 1.95×10^{-12} , respectively; the exact inputs and step are retained in results/born-permittivity-study/component-permittivity-derivative-check.json.

11.1.2. Sparse structure screen

The first screen used 12 retained states: low, middle, and high loading at the two best-covered pressure temperatures (40 and 60 °C) and the two best-covered speciation temperatures (20 and 40 °C). All five permittivity rules were crossed with original Born and SSM+DS. Table 5 reports evaluated attempts and positive-target log₁₀ RMSE. The attempt count is a solver diagnostic, not an acceptance cutoff or scientific ranking coordinate.

Variant	States	ϵ_r range	p_{CO_2} RMSE	Pressure factor	Species RMSE
A original	12/12	57.13–79.29	0.2704	1.4008	0.0551
A SSM+DS	10/12	57.13–79.08	0.4910	2.7414	0.1358
B original	12/12	57.98–74.39	0.5017	2.6384	0.0728
B SSM+DS	9/12	57.57–74.28	0.5742	4.3025	0.1451
C original	12/12	43.03–63.78	0.9231	8.2675	0.0761
C SSM+DS	12/12	42.55–63.73	0.4020	2.4929	0.1317
D original	12/12	56.51–70.73	0.4645	2.8418	0.0514
D SSM+DS	11/12	56.50–69.66	0.5251	3.8518	0.1193
E original	11/12	34.63–60.22	1.3375	26.3051	0.0299
E SSM+DS	12/12	32.68–60.03	0.4669	2.2065	0.0960

SSM+DS does not improve every dielectric rule. It increases both errors for A, B, and D; it reduces the pressure RMSE by 0.5211 for C while increasing the species RMSE by 0.0555; and it reduces the E-rule pressure RMSE by 0.8705, but increases the species RMSE by 0.0660. Thus the apparent rescue of the Figiel package is primarily the corrected SSM+DS radial work, whereas the large baseline differences among A–E arise from the permittivity rule itself. This interaction rules out a single independent “Born correction” effect.

Because E-SSM+DS completed the sparse screen, the permitted follow-up tested universal 7.01 against Zuber-derived analog/fallback α_i , and $f_{\text{solv,MEA}} = 1.0, 1.4, 1.5, 1.6$. The 1.5 sparse case lost two states despite its smaller pressure RMSE; 1.4 and 1.6 completed all 12. An exact post-solve pool crossover at the 12 converged 1.6 compositions found $x_{\text{CO}_2} \leq 0.002918$, $|\Delta\epsilon_r| \leq 0.2663$ (0.813%), and $|\Delta f_{\text{mix}}| \leq 0.001249$ when CO₂ was excluded. That algebraic result was

used only as a cheap preliminary screen. The promoted comparison below then re-solved every state with molecular CO_2 removed from both pools so pressure, speciation, phase identity, and convergence were tested in the coupled system.

11.1.3. Complete retained-state comparison

The promoted configurations attempted all 161 active pressure observations from 313.15 to 393.15 K and all 44 speciation states from 293.15 to 353.15 K: 1,640 configuration–state attempts in total. The candidate Engine wheel preserves the known-state result to numerical precision and reduces that solve from 12.375 to 1.472 s (8.41 times). This parity timing was measured on the immediate predecessor candidate at Engine commit 355d950; the complete comparison used the final d8e02e4 immutable study wheel, which adds explicit neutral-pool exclusion, its tested derivatives, and preservation of final partial evidence at the unchanged bubble-iteration limit. The same solver path, tolerances, and starts were used with 12 workers. The faster wheel remains installed; no scientific hierarchy is inferred from whether an Engine commit has merged to main.

The Engine’s fixed-pressure density anchor was measured separately at v1e_obs_0140. Eight single-threaded repetitions reduced median Mixture.state time from 58.98 to 14.76 ms and density iterations from 30 to 1 (4.00 times), with zero pressure or density difference. The present reactive equilibrium callable owns its density closure and does not expose this anchor, so the measurement did not alter the formulation campaign.

The separate 319-row cross-validation inventory remains non-executable: all 319 rows lack a residual scale and immutable data packet, and the 198 speciation rows also lack a fixed-pressure definition. The 123-state immutable packet used here is already executable and is not evidence that those broader admission decisions are resolved.

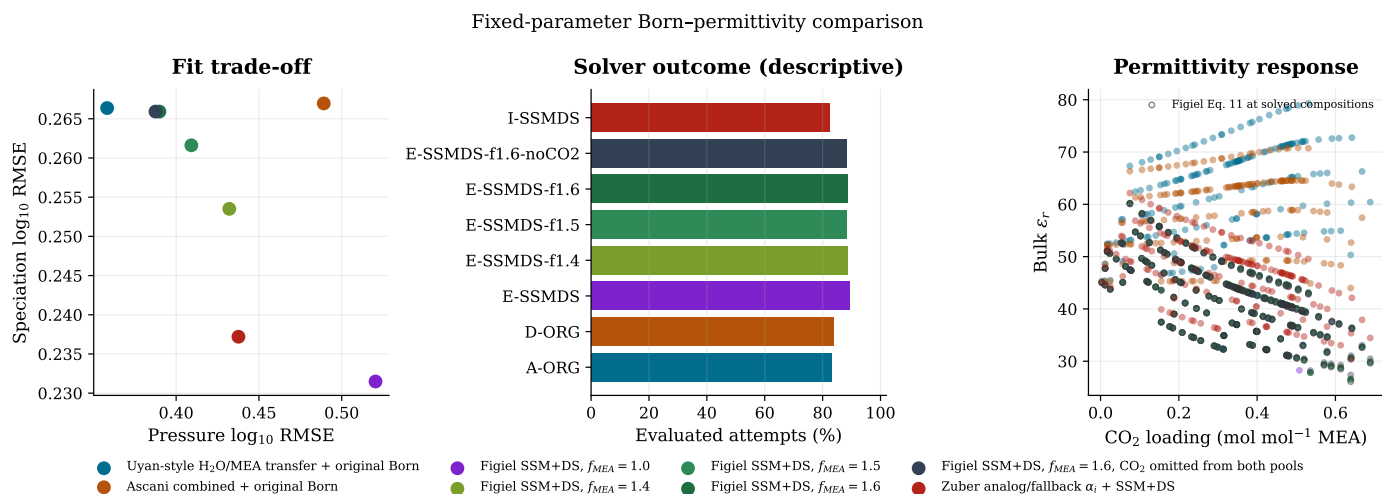


Figure 3: Fixed-parameter trade-off, descriptive evaluated-attempt fraction, and bulk relative permittivity for every promoted configuration. The open circles apply Figiel’s Eq. 11 to A-original solved compositions; they are a crossover, not independent loaded-MEA dielectric observations. Attempt fraction is not used to rank formulations.

Variant	States	p_{CO_2} RMSE	Pressure factor	Mean bias	Species RMSE
A original	170/205	0.3583	1.9577	0.0956	0.2664
D original	172/205	0.4892	2.5954	0.2763	0.2670
E SSM+DS, $f_{\text{MEA}} = 1.0$	183/205	0.5204	2.7461	0.2372	0.2315
E SSM+DS, $f_{\text{MEA}} = 1.4$	182/205	0.4321	2.2891	0.2177	0.2535
E SSM+DS, $f_{\text{MEA}} = 1.5$	181/205	0.4092	2.1997	0.2114	0.2616
E SSM+DS, $f_{\text{MEA}} = 1.6$	182/205	0.3899	2.1264	0.2074	0.2659
E SSM+DS, $f_{\text{MEA}} = 1.6$, no CO_2	181/205	0.3875	2.1090	0.2066	0.2659
Zuber analog/fallback α_i SSM+DS	169/205	0.4376	2.3821	0.0710	0.2372

Because different configurations fail on different rows, those full-row statistics are not directly paired. On the exact positive-target intersection shared by all eight configurations ($n = 102$ pressure and $n = 129$ speciation), A-original still has the smallest pressure log RMSE (0.3544; median factor 1.8607), while E-SSM+DS with $f_{\text{MEA}} = 1.0$ has the smallest speciation log RMSE (0.2333). The complete paired table is retained in `results/born-permittivity-study/full-common-row-statistics.csv`.

Figure 4 shows that every configuration shares the same principal pressure pattern: negative low-loading residuals, positive middle-loading residuals, and a temperature-dependent high-loading return. The speciation statistic is dominated by underprediction of HCO_3^- ; the other reported quantities cluster near zero residual. The grouped heat map makes the temperature, source, and species dependence explicit.

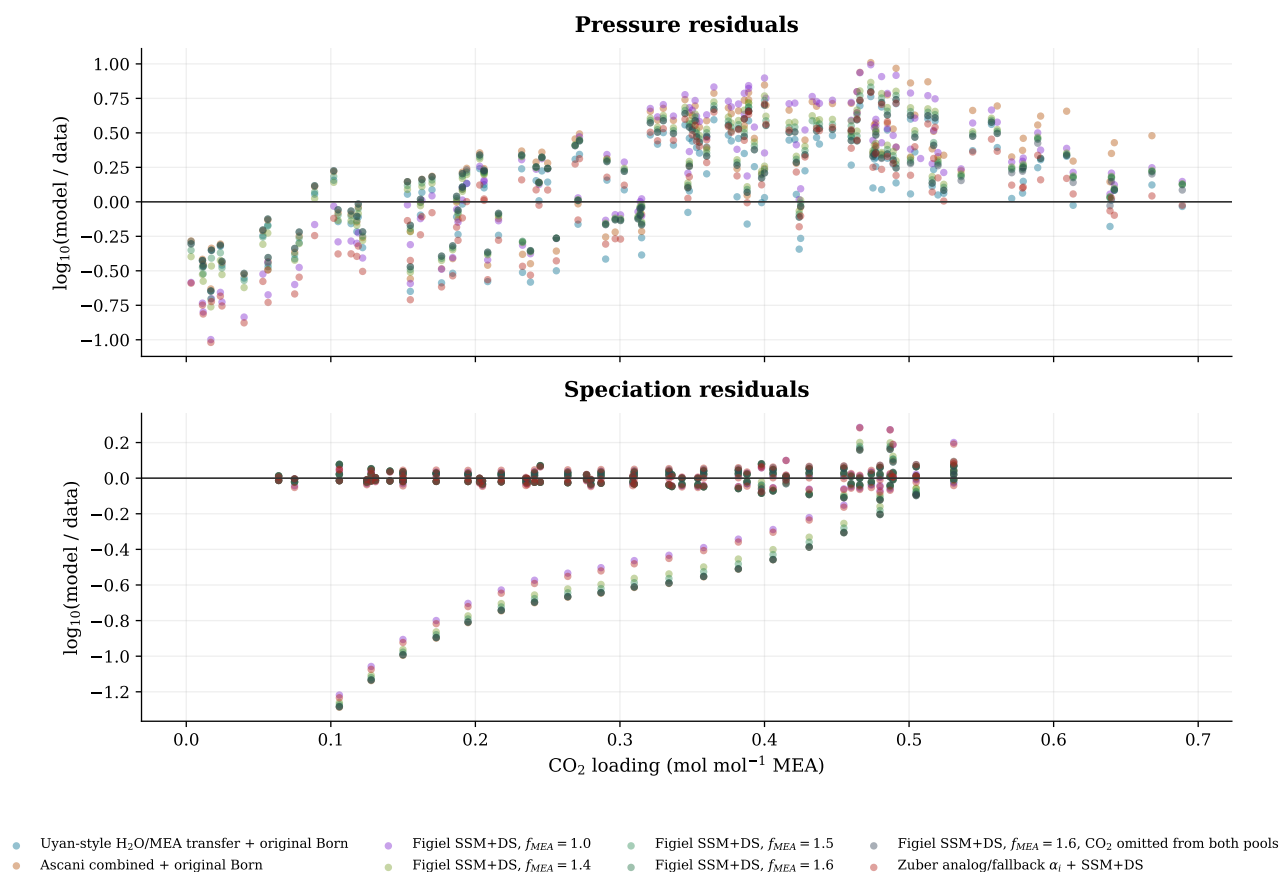


Figure 4: All positive observed–predicted pressure and speciation residuals. Every point is a direct retained Engine evaluation; no curve or failed state has been interpolated.

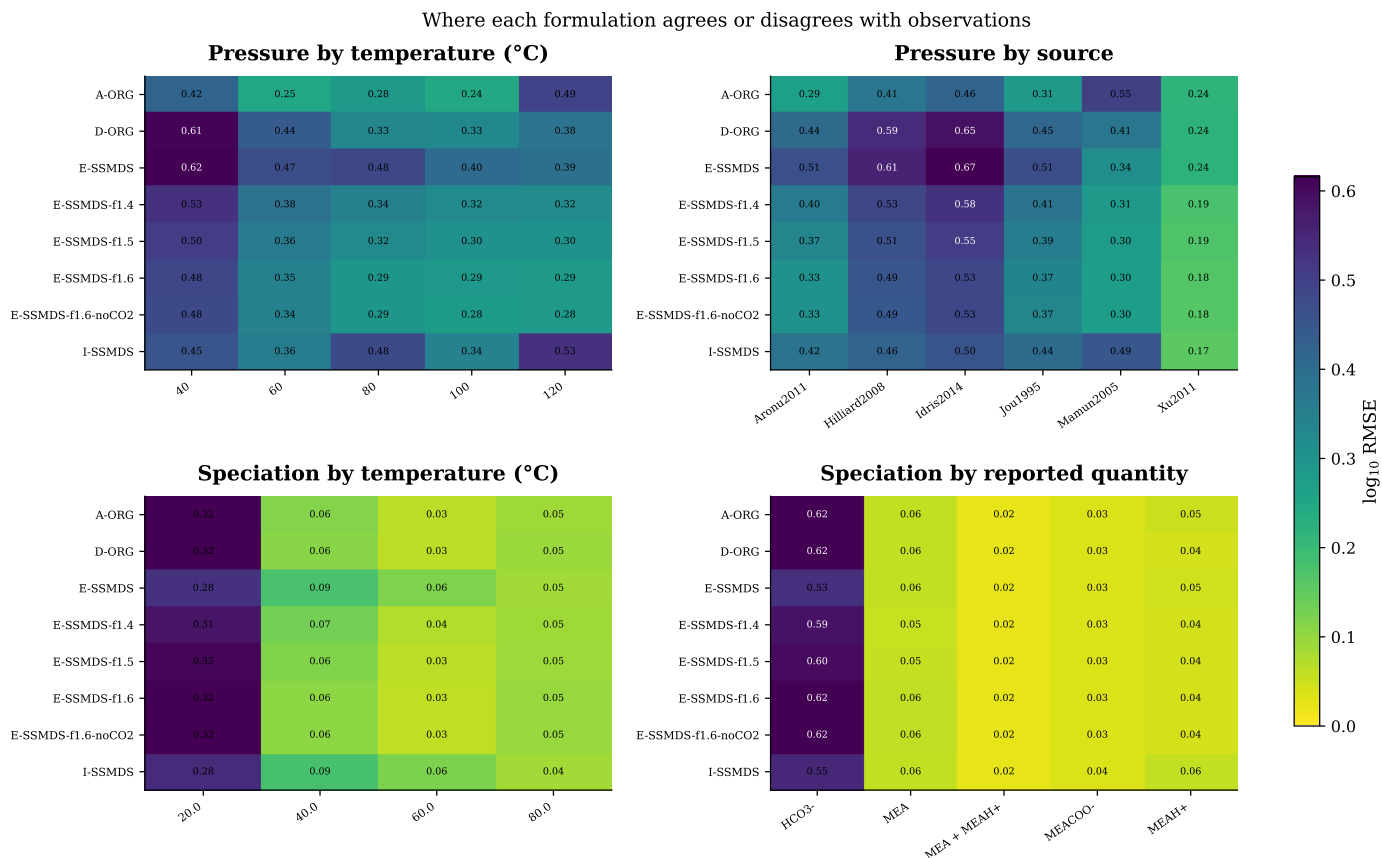


Figure 5: Log₁₀ RMSE grouped by pressure temperature and source, and by speciation temperature and reported quantity. Missing groups remain missing rather than being imputed.

Every evaluated liquid state is on an accepted liquid mechanical branch, and every incipient vapor is on its accepted vapor branch. Maximum absolute charge closure is below 2.4×10^{-17} , material closure below 2.8×10^{-13} , reaction-affinity closure below 1.3×10^{-11} , and relative pressure closure below 6.4×10^{-10} . Figure 6 retains the density range, closure values, and all 220 non-evaluatable attempts. Of those, 181 reach the existing bubble-pressure Newton iteration limit, 38 produce a finite phase candidate that does not pass the complete equilibrium certificate, and one stops in the inner IPOPT solve. The complete row-by-row table records source, temperature, loading, continuation and start availability, and stopping point. The former label “request-policy validation” was a reporting defect: an empty policy record hid the native “bubble Newton iteration limit reached” result. Engine commit d8e02e4 preserves the last partial state and policy so that the actual failure is now retained. These are execution boundaries, not evidence that a physical formulation is false.

Source	Temperature (°C)	Bubble iteration	Certificate	Inner solve
Aronu 2011	40, 60, 80	0	14	0
Böttinger 2008	20	0	1	0
Hilliard 2008	40	0	13	1
Idris 2014	40	0	1	0
Jou 1995	40, 60, 80, 120	42	5	0
Mamun 2005	120	70	1	0
Xu 2011	100, 120	69	3	0
Total		181	38	1

These counts are formulation–state attempts over 60 distinct observations. All failures had a continuation value. Of the

181 bubble-iteration stops, 165 used one pressure start without a liquid start and 16 also had a liquid start. Of the 39 later stops, 38 used one pressure start and one certification failure had no pressure start. The exact loading and per-variant record remains in `results/born-permittivity-study/full-failure-characterization.csv`.

All 38 certificate failures compiled, completed the inner solve, passed the EOS-domain check, and returned a locally evaluable candidate, but no candidate passed the full closure and local-stability certificate. Thirty-seven are reported as numerical-convergence failures; one E-SSM+DS, $f_{\text{MEA}} = 1.5$ attempt at 20 °C is a physical-rejection classification with material closure 0.0243, reaction-affinity closure 1.893, KKT closure 0.0125, and failed local stability. The same observation evaluates under the other seven configurations. The single IPOPT failure likewise occurs for one configuration of a row evaluated by most alternatives. These facts make all 39 candidate/inner-solve failures branch-and-initialization robustness evidence, not proof that the observed state or formulation is physically impossible.

The 181 bubble failures are concentrated at high temperature: 182 of all 220 non-evaluable configuration-state attempts occur at 120 °C, with the remainder at 20, 40, 60, 80, and 100 °C. Increasing the outer iteration cap from 18 to 60 and sweeping pressure starts from 10 kPa to 5 MPa did not recover the representative 120 °C row; iterates alternated between gas-like and dense-liquid roots. The required Engine experiment is therefore a branch-aware coupled pressure KKT formulation, not looser tolerances or more iterations. Every one of these 181 rows is retained as an acceptance case: the structural comparison must be rerun after that solver change, and no present attempt count is used to rank formulations.

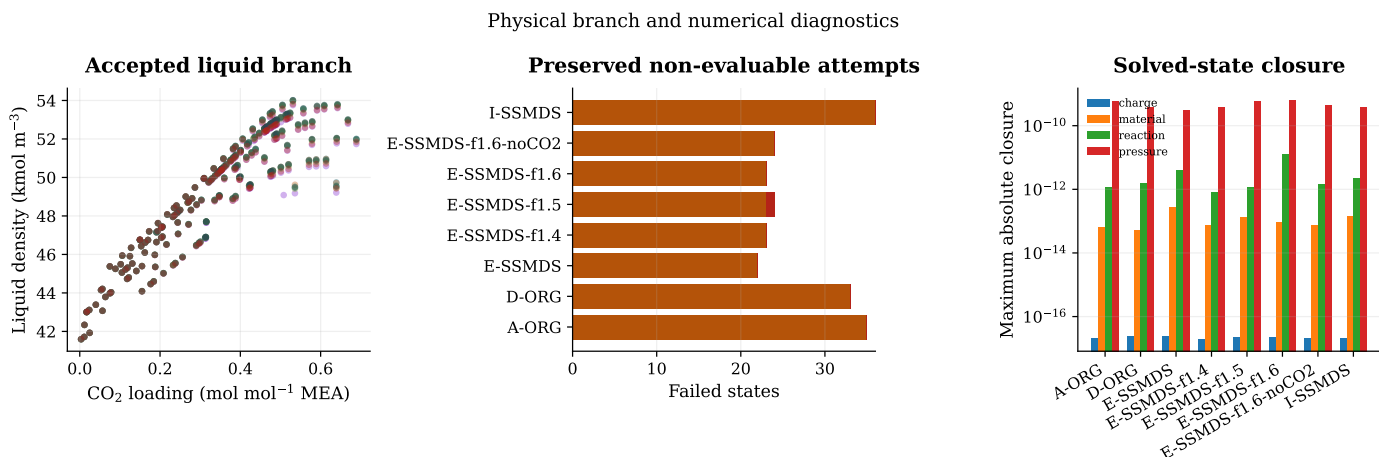


Figure 6: Accepted liquid density, preserved failure classes, and maximum charge, material, reaction-affinity, and pressure closure for every promoted configuration.

At the common 40 °C, loading 0.389 state, A-original has $(a^{\text{Born}}, a^{\text{DH}}, a^{\text{res}})/(RT) = (-13.8158, -0.0577, -22.4172)$; E-SSM+DS with $f_{\text{MEA}} = 1.6$ gives $(-12.7242, -0.0989, -21.3758)$. The separated axes in Figure 7 expose the otherwise hidden Debye-Hückel change.

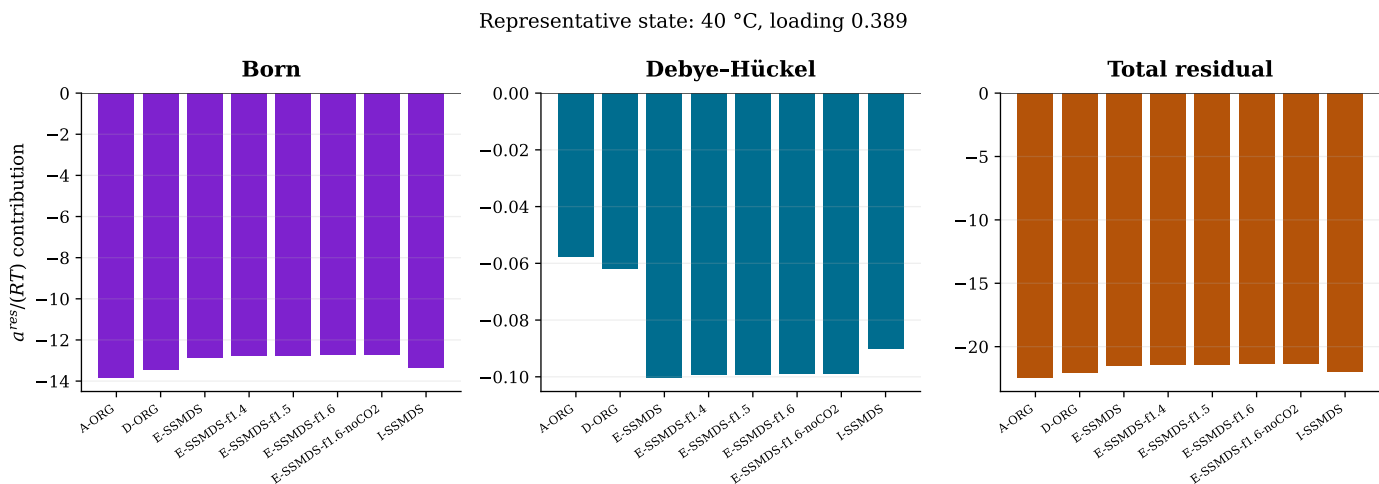


Figure 7: Born, Debye-Hückel, and total residual Helmholtz contributions at one common representative state. Each panel has its own ordinate scale.

The retained dielectric folder contains no admitted static-permittivity data for loaded aqueous MEA. The plotted Figiel relation is therefore a literature-form crossover evaluated at this study’s solved compositions, not independent MEA validation. Within that limit, E predicts 26.54–60.19, the Zuber analog/fallback suppression screen predicts 30.36–62.15, D predicts 44.00–70.80, and A predicts 45.32–79.29 across successful states. Coupled molecular- CO_2 exclusion from both f_{mix} and the neutral dielectric pool changes the universal-1.6 pressure log RMSE from 0.389898 to 0.387514, changes species log RMSE by less than 2×10^{-7} , and has negligible effect on the present paired residuals.

decision: no alternative simultaneously improves pressure, speciation, dielectric plausibility, and branch identity on the evaluated fixed-parameter evidence. Retain A-original (Uyan-style $\text{H}_2\text{O}/\text{MEA}$ transfer with original Born) in the active bundle and do not change the handoff ZIP. E-SSM+DS at $f_{\text{MEA}} = 1.0$ is the best species-error alternative, while $f_{\text{MEA}} = 1.6$ is the best pressure member of the universal-Figiel family; neither improves the current pressure result. Explicit CO_2 exclusion is not material at the present compositions. The next discriminating experiment is the Engine pressure-in-KKT robustness correction using all 181 preserved bubble failures as acceptance evidence, followed by a complete rerun and independent loaded-MEA static-permittivity comparison. Broad parameter fitting remains deferred until that structural comparison has complete row behavior.

12. Current conclusions and next experiments

The selected physical structure includes reciprocal induced $\text{CO}_2\text{-H}_2\text{O}$ association and excludes CO_2 self-association. The active foundation uses corrected R1–R3, the Uyan-style solvent-only dielectric transfer, and $k_{\text{CO}_2, \text{MEA}} = 0$. The following steps provide the shortest discriminating path. Items below are proposed work, not retained results.

1. Diagnose the remaining temperature sign change using the retained row-level residuals. Test a held-out-source or held-out-temperature replay before changing another coordinate. Do not extend the R4 slope beyond the tested bound: that direction ceased improving the screen and began losing speciation evaluations.
2. Complete the remaining coupled bubble-pressure ablation with the active parameter file: nine versus six liquid species, crossed with ePC-SAFT versus ideal vapor. The retained comparison has already established the exact six-species reaction projection, ideal-liquid behavior, fixed- T, P liquid behavior, and the failure of the neutral apparent-flash shortcut. Report state evaluability, pressure and species metrics, charge/balance closure, and changes by pressure range and temperature.
3. Admit independent physical-solubility evidence before expanding the model. Jiru et al. measure physical CO_2 solubility in MEA–water by the N_2O analogy over 298.15–323.15 K [13]; Hartono et al. report N_2O solubility in unloaded and loaded MEA [10]. Wong et al. directly measure dissolved molecular CO_2 and Henry behavior [20],

but every retained row contains $0.517 \text{ mol kg}^{-1} \text{ NaClO}_4$, which the nine-species model omits; use those rows as an external challenge unless the salt is represented explicitly.

4. Acquire or digitize independent static-permittivity observations for loaded aqueous MEA. The completed fixed-parameter comparison cannot select between the A and E dielectric responses from pressure and speciation alone, and the Figiel relation is not an independent MEA measurement. Keep A-original as the baseline until that evidence exists.
5. With R1–R5, the ion block, and independently accepted $k_{\text{CO}_2, \text{H}_2\text{O}}$ fixed, compare only physical dissolved- CO_2 alternatives that the new data can distinguish: a CO_2 –MEA association edge, a physically identified solvation/activity contribution, and any richer binary correction justified by the temperature and composition coverage. Do not combine coordinates that the admitted observations cannot identify.
6. Require simultaneous comparative improvement in pressure, speciation, and convergence coverage, followed by held-out temperatures and sources before issuing an immutable packet. No unapproved numerical threshold is an acceptance gate.

unknown: whether ideal vapor, reduced chemistry, or neither materially changes the coupled bubble-pressure residual has not been tested with the active bundle.

Replace the current numerical foundation in place when a better MEA-owned regression or validation result passes the stated comparison gates. Every replacement must bind the Engine wheel and hash, parameter identities and evidence status, admitted rows and hashes, optimizer diagnostics, validation summaries, and the transfer decision. This PDF is the live calculation and interpretation record for the active bundle, including the reported fit statistics and state counts.

12.1. Independent review record

An independent read-only review on 2 September 2026 assessed commit fb6f948. It verified the 161-pressure plus 44-speciation scope, the coupled CO_2 -pool exclusion, normalized f_{mix} , separate source hashes, Rules B/C derivative checks, preserved failures, recomputed summary metrics, and the synchronized 27-page render. The verdict was *Review Pass*: no material scientific, numerical, provenance, or layout finding remained. The approved use is fixed-parameter model-structure selection and failure characterization. It does not promote a new active bundle, resolve the separate 319-row admission decisions, or supply missing independent loaded-MEA static-permittivity observations.

13. Provenance

The base numerical reconstruction uses the PR #134 file:

`validation/campaigns/2026-mea-parameter-estimation/analysis/joint-fit/parameters.json`

at commit 38e91823b6d4f26c1d549f07aaef24a089d8e16d. The selected live mapping is:

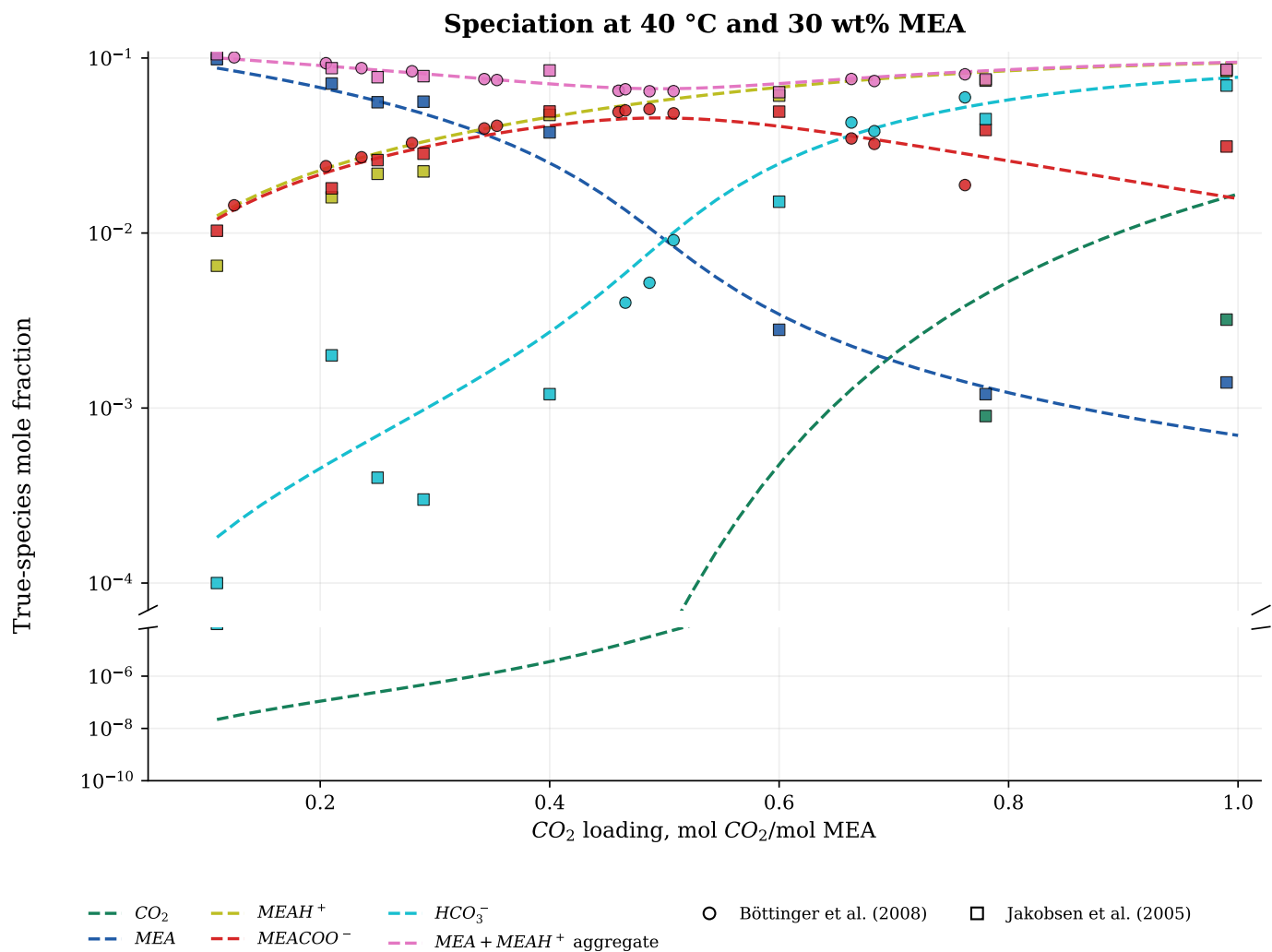
`results/selected-current-best-parameters.json`

Its SHA-256 is the parameter identity reported in Section 2. The comparison result records both the base and selected fingerprints.

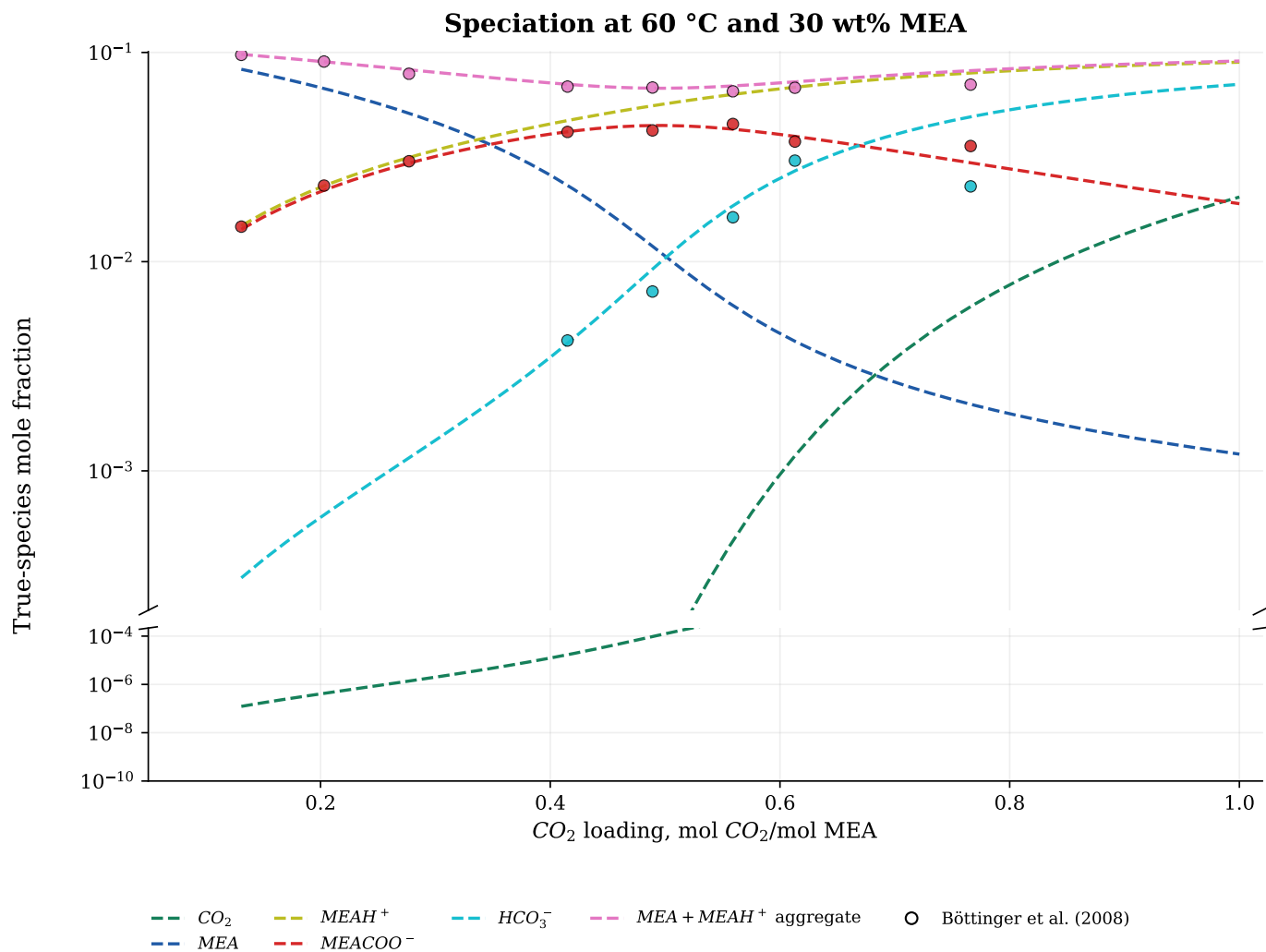
Source	Role
ePC-SAFT Issue #80	Upstream full regression and immutable-packet campaign.
ePC-SAFT Issue #119	Bounded refinement scope and current foundation reconciliation.
ePC-SAFT PR #134	Closed, unmerged blocker-evidence change. It produced no parameter bundle; this notebook reconstructs only its frozen foundation vector.
Local component source audit	docs/ePC-SAFT/full-component-parameter-source-audit.md; species-level lineage and evidence gaps.
Local model-hierarchy review	docs/ePC-SAFT/amine-epcsaft-model-hierarchy-literature-review.md; selected nine-species structure and induced-association decision.
Retained parameter evidence	analyses/reactive_epcsaft_parameter_evidence/; supported-negative and diagnostic evidence.
Retained local-speciation fit	Git commit 897bb6e; the May 2026 8-state/22-residual principal-ion fit retained as Class 3 evidence awaiting promotion checks.
Reaction exporter defect	validation/campaigns/2026-mea-parameter-estimation/scripts/build_fit_input.py at Engine commit 38e91823b6d4f26c1d549f07aaef24a089d8e16d.
Corrected start comparison	results/parameter-start-comparison.csv; full corrected-foundation and Böttinger-R4 comparison statistics.
Bounded sensitivity screen	results/parameter-sensitivity-screen.csv; common-row CO ₂ -MEA and CO ₂ -water perturbation results used in Section 7.
Quick direct endpoint screen	results/quick-endpoint-perturbation-screen.csv and its JSON receipt; full reactive bubble-point endpoints and the bounded SciPy reaction-root spot check.
Best-in-slot campaign	results/best-in-slot-campaign/; centered sensitivity, R4 pivot-slope grids, full validations, boundary diagnostics, exact receipts, and the selected candidate's row-level Engine evaluations.
Born-permittivity structure study	results/born-permittivity-study/; exact sparse and complete state/target tables, failure characterization, grouped statistics, Engine parity and density-anchor timings, deterministic formulation construction, and hashes. Figures 3–7 read these retained values without rerunning the Engine.
Candidate Engine wheel	data/input/engine-candidates/406cd4e/; Engine commit 406cd4e942a1e96260debafdc23c4b8baf7c6226 and wheel SHA-256 529051a41e5fe2e3a8f944500d1fc2ddf4deaf57cc54f151d1d68d13b6f67a4a. The 8.41-times parity timing was measured on immediate predecessor 355d950; the physical result was unchanged.
Selected live mapping	results/selected-current-best-parameters.json; Uyan-style H ₂ O/MEA transfer with the selected Class 3 R4-CO ₂ fit.
Trace-species abundance	figures/speciation/output/speciation-model.csv; retained 44-state calculated values summarized in Section 10.
Current IDAES comparison	Official public MEA solvent and MEA vapor property files; used only to define the proposed structural ablation.
Matched species comparison	../enrtl_six_species_ideal_comparison/notebook.qmd and ../enrtl_six_species_ideal_comparison/notebook.pdf; exact ideal projection, fixed- <i>T</i> , <i>P</i> Engine/SciPy comparison, corrected apparent-pressure diagnostic, parameter inventory, sensitivity survey, UQ checkpoints, and figures.

A. Additional-temperature speciation replays

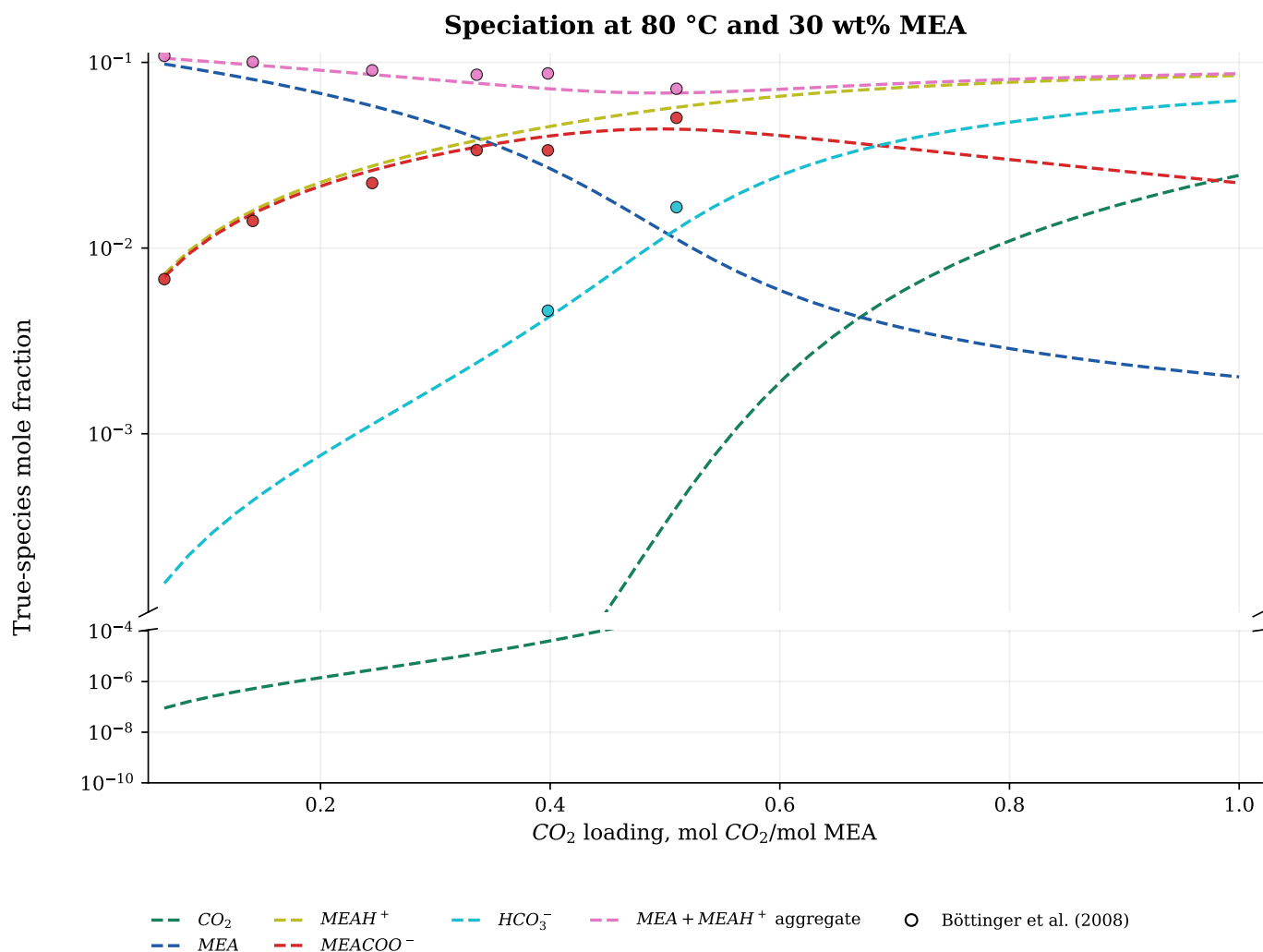
The following panels apply the same species selection, source encoding, reported-zero omission, split-axis rule, parameter bundle, and pinned Engine route as Figure 1. They are supplementary temperature views rather than additional fits. At 40 °C, 74 positive plotted values cover 21 states and observed loading 0.11–0.99; at 60 °C, 21 values cover eight states and observed loading 0.131–0.766; at 80 °C, 14 values cover six states and observed loading 0.064–0.51. Every model grid continues to loading 1.0, and all 46 states evaluated at each temperature. Dashed lines connect those direct calculations.



Appendix Figure A1. Principal-species replay at 40 °C and 30 wt% MEA. Symbols retain every displayed Böttinger and Jakobsen observation; dashed lines are direct Engine calculations.



Appendix Figure A2. Principal-species replay at 60 °C and 30 wt% MEA. Symbols retain every displayed Böttinger observation; dashed lines are direct Engine calculations.



Appendix Figure A3. Principal-species replay at 80 °C and 30 wt% MEA. Symbols retain every displayed Böttinger observation; dashed lines are direct Engine calculations.

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